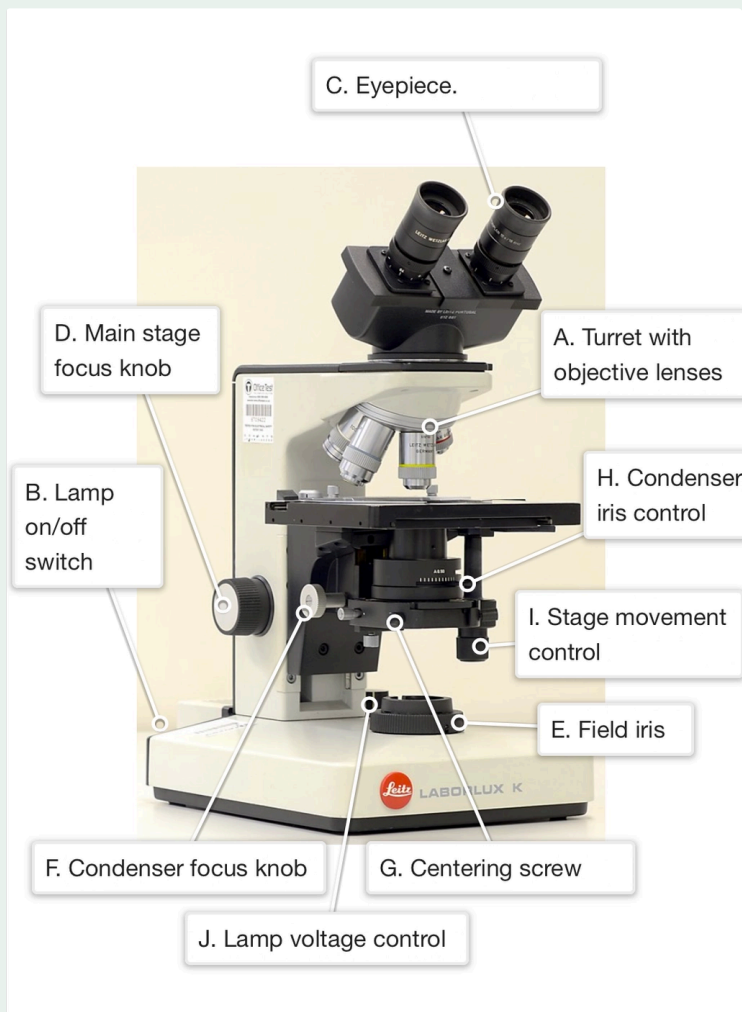


3.6 Diagnostic Role of the Blood Film

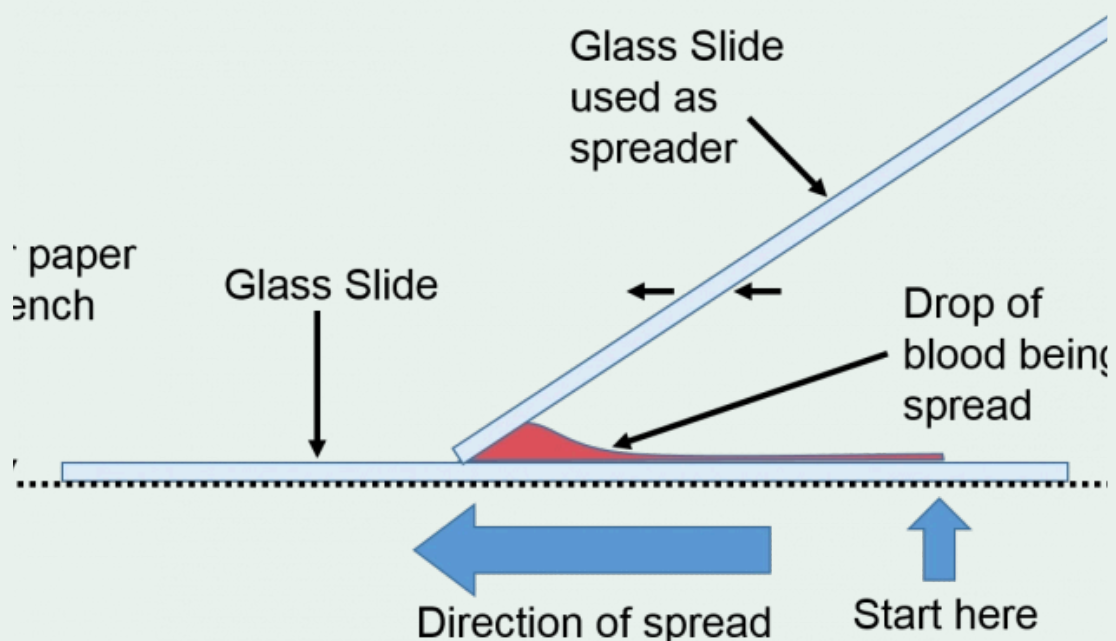
- ▼ Which stain, used in this practical, is the most frequently used stain for visualisation under microscopes?
 - Haematoxylin & Eosin (H&E)
 - This stain is used to view cells at visible wavelengths
- ▼ What is the role of haematoxylin in the stain?
 - It is blue, **binds DNA** and **shows up the nucleus**.
- ▼ What is the role of eosin in the stain?
 - It is pink, **binds protein components**, particularly in the **cytoplasm**
- ▼ Image showing a typical light microscope.



▼ Describe the steps involved in setting up a microscope correctly.

- Preparation: Make sure the **microscope is set to the lowest magnification (4x)** and **ensure that the lamp is turned on**
- **Step 1: Focus the image**
 - Use the main stage focus knob to make sure the image is in the right focus to see it clearly
- **Step 2: Focus the condenser**
 - Close the field iris and adjust the condenser focus knob until you get a sharp image of the edge of the disc of light coming through the iris diaphragm
 - This should be roughly in the centre of the field of view – if it is not, adjust it with the centering screws
- **Step 3: Adjust the field iris**
 - Open the field iris again and stop when the whole field of view is only just illuminated.
- **Step 4: Adjust the condenser iris**
 - Start with the condenser iris fully open and then close it until it only just begins to darken the image.
- **Examine:** Look at your specimen, **making any fine focus adjustments where necessary**

▼ Image showing how to spread blood across a microscope slide correctly.



▼ What is the function of the field iris of a microscope?

- To **control the size of the "field" or area** that you are viewing under a microscope

▼ What is the function of the condenser of a microscope?

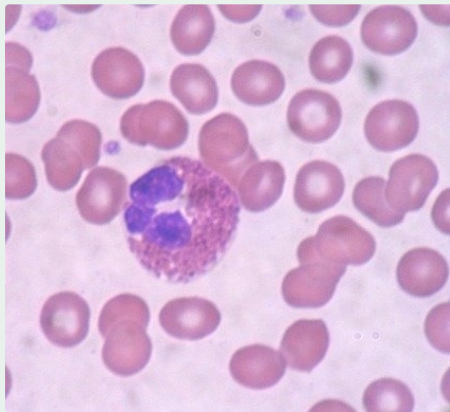
- To allow a **uniform and maximum intensity of light** to be transferred **from the lamp to the area of the slide (field) that you are examining**

▼ What is special about the staining of white blood cells under Leishmann's stain?

- **White blood cells actually stain blue**

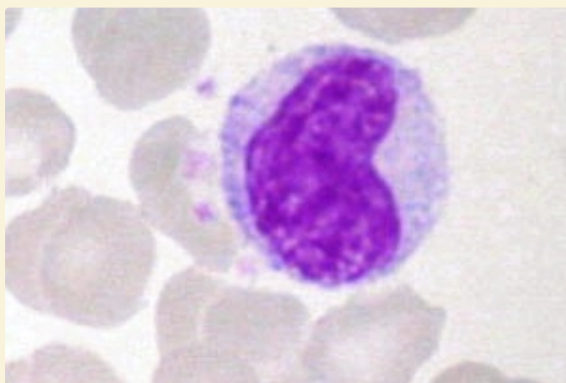
▼ What properties are characteristic of eosinophils in identification under a microscope?

- The bi-lobed nucleus and pink granules



▼ What property is characteristic of monocytes in identification under a microscope?

- The **indented nucleus** is a good way of identifying a leukocyte as a monocyte.



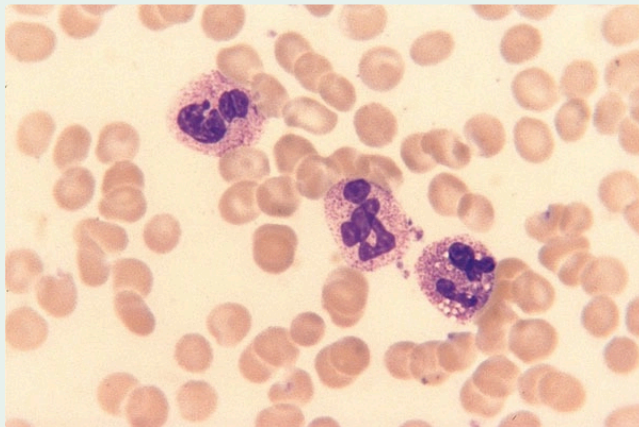
- ▼ What properties are characteristic of basophils in identification under a microscope?
 - The **blue stained granules** and **lobed nucleus** mark a leukocyte as a basophil.
- ▼ What properties are characteristic of lymphocytes in identification under a microscope?
 - The **large round nucleus** and **relative lack of cytoplasm** marks a cell as a lymphocyte.
- ▼ What can happen to the cytology of neutrophils after a severe bacterial infection?
 - The presence of **clear vacuoles within the cytoplasm** of the neutrophils can be seen
- ▼ What general chromosomal abnormality needs to occur for chronic myeloid leukaemia (CML) to be present?
 - **Translocation** on the **Philadelphia chromosome**
 - i.e abnormal chromosome 22
 - This causes the **BCR:ABL1 fusion gene** behind CML
- ▼ What is megaloblastic anaemia in terms of leukocytes?
 - Anaemia where the **nucleus is retarded (behind in development) in comparison to the cytoplasm**

▼ Image showing the different blood cells present in a blood film.

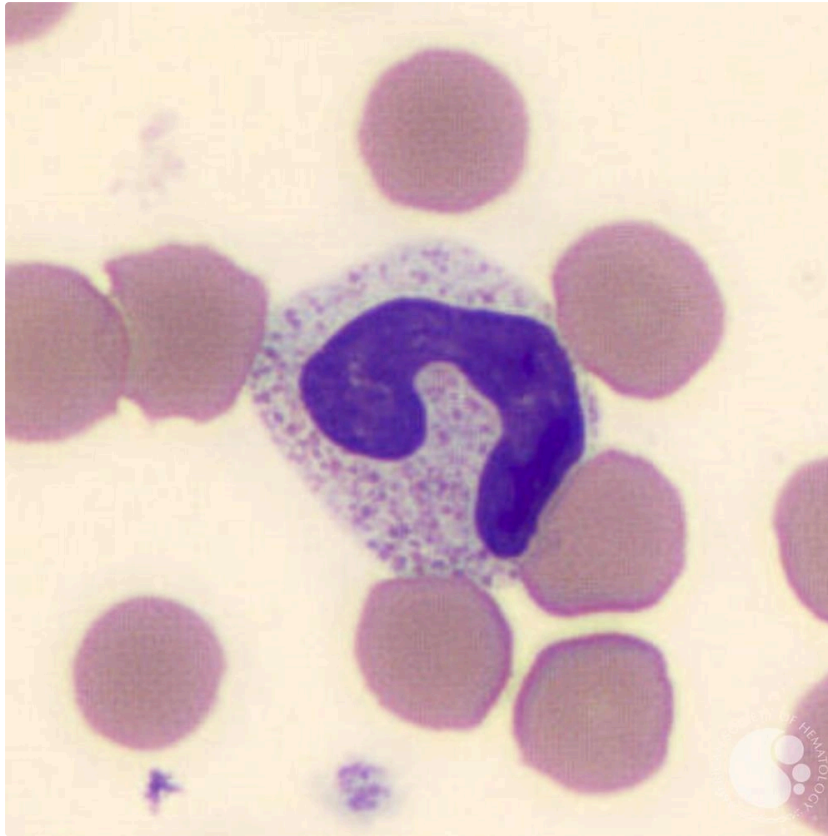
Non-nucleated cells (their precursors in the blood marrow had nuclei, but these were lost before they reached the blood stream)			
Red blood cell (erythrocyte) Most abundant blood cell, involved in O ₂ transport. Biconcave shape, dusky pink. 7 µm diameter.		Blood platelets These are important in blood clotting, and will probably be found in small clumps on a smear. Bluish. 2-4 µm diameter.	
Nucleated cells (also known as white blood cells or leukocytes, easily identified by their blue staining nuclei)			
Lymphocyte Small cells with near spherical nucleus and little cytoplasm. Involved in the immune response. Around 8 µm diameter.		Eosinophil Cytoplasmic granules are bright pink. Bi-lobed nucleus. Implicated in fighting parasitic infections and allergy. Around 12 µm diameter.	
Neutrophil Multi-lobed nucleus ("polymorphonuclear"), cytoplasmic granules. Phagocytic cells which kill bacteria. 10-12 µm diameter.		Basophil Cytoplasmic granules staining strongly blue. Lobed nucleus. Involved in inflammation and allergic responses. 10-15 µm diameter	
Monocyte Indented nucleus and pale granules in the cytoplasm. Phagocytic cells with a role in the immune response and inflammation. 17-20 µm diameter.			

▼ What does being able to see clear vacuoles through a neutrophil suggest?

- Bacterial infection



- ▼ Image showing a band form neutrophil (nucleus has not yet lobulated).



- ▼ What genetic feature is essential in chronic myeloid leukaemia (CML)?
- **BCR-ABL1 fusion gene**
 - Its presence can signify CML
- ▼ What sign in leukocytes can help us to identify megaloblastic anaemia?
- **Hypersegmented nuclei** of neutrophils